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TITLE PAGE

- **Problem Statement ID –** SIH26003
- **Problem Statement Title-** AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)
- **Theme-** MedTech / BioTech / HealthTech
- **PS Category-** Software
- **Team ID-** LIH143
- **Team Name-** Sentinel



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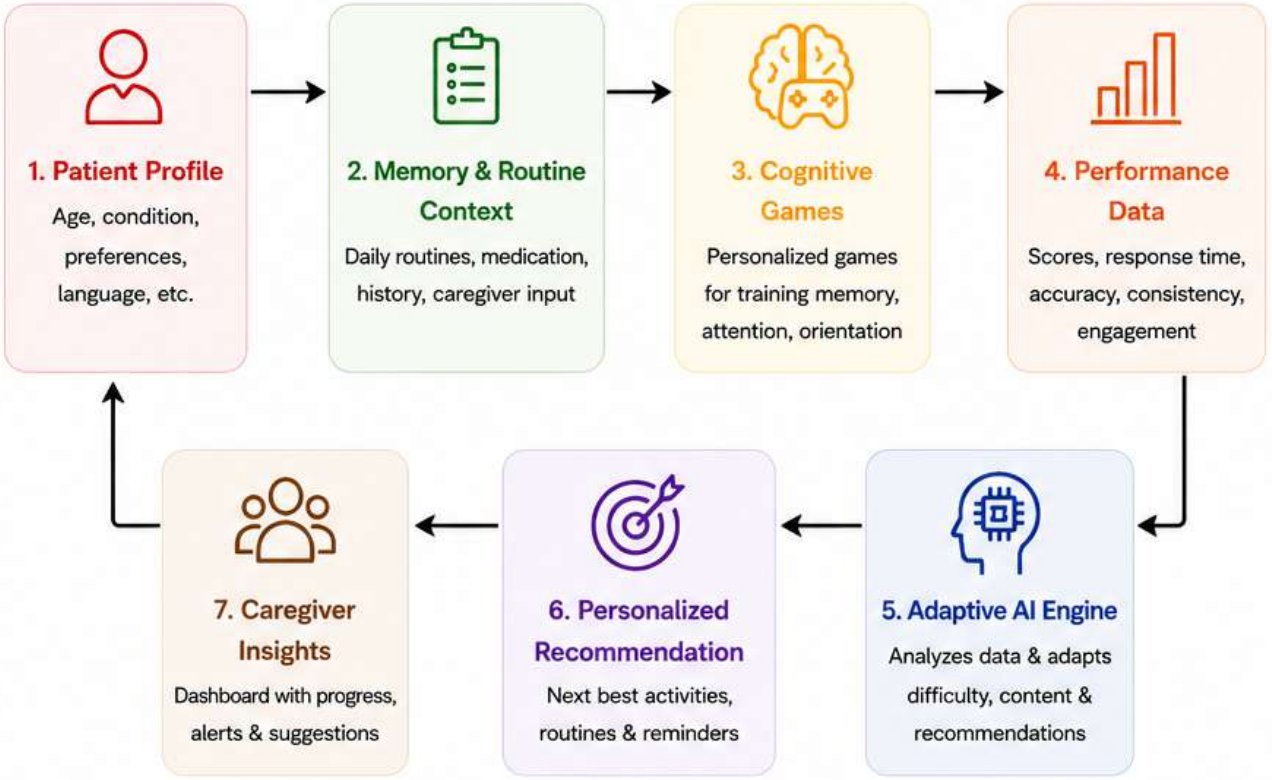
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PROPOSED SOLUTION

CogniCare is an AI-powered cognitive gaming and memory assistance platform that provides personalized cognitive training, daily living reminders, and caregiver insights to improve the quality of life for elderly dementia patients in NER.

SOLUTION ARCHITECTURE & WORKFLOW



KEY FEATURES

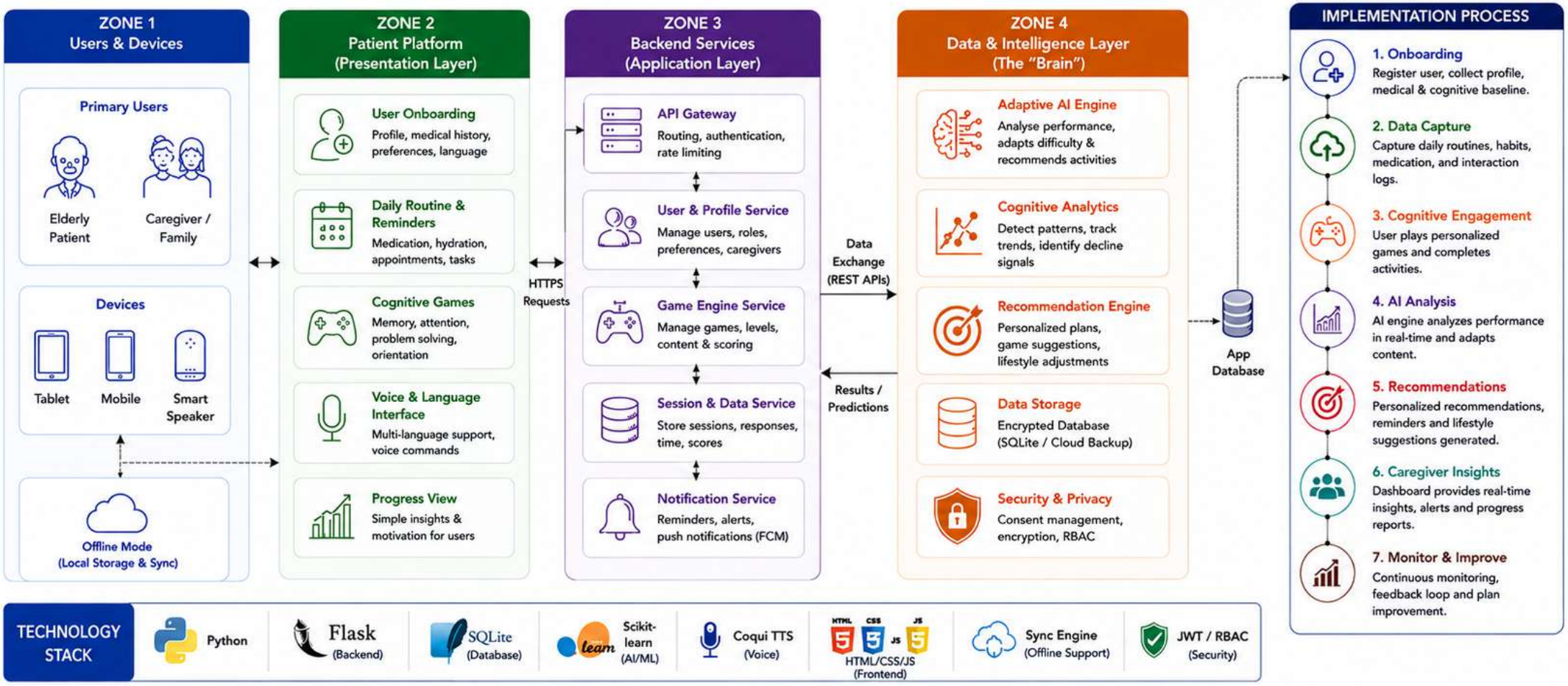
- ✓ **AI-Powered Cognitive Training:** Adaptive games based on individual performance.
- ✓ **Daily Living Reminders:** Medication, hydration, appointments & task reminders.
- ✓ **Multi-language & Voice Interface:** Local language support for better accessibility.
- ✓ **Caregiver Dashboard:** Real-time insights, progress tracking & smart alerts.
- ✓ **Offline-First Design:** Works in low connectivity areas of NER.
- ✓ **Secure & Privacy-Focused:** Data protection with consent and role-based access.

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TECHNICAL APPROACH

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TECHNOLOGY STACK

Python | Flask (Backend) | SQLite (Database) | Scikit-learn (AI/ML) | Coqui TTS (Voice) | HTML/CSS/JS (Frontend) | Sync Engine (Offline Support) | JWT / RBAC (Security)

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FEASIBILITY AND VIABILITY

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FEASIBILITY

- Uses proven AI/ML models for adaptive cognitive training and performance analysis.
- All user data is securely stored and processed using cloud with local fallback.
- Voice, multilingual support and offline mode ensure accessibility in NER.
- Modular architecture enables easy development, testing and future scaling.

VIABILITY

- Strong need for dementia care solutions in India, especially in underserved regions.
- Low-cost, high-impact solution with real social and caregiver value.
- Scalable model with potential partnerships (NGOs, Hospitals, Govt. programs).
- Sustainable through freemium model and institutional collaborations.

PRACTICAL IMPLEMENTATION

- MVP ready and pilot rollout in care facilities within 6 to 9 months.
- Works offline with local storage and sync when internet is available.
- Simple and intuitive UI for elderly users and caregivers.
- Built using accessible technologies and open-source tools for faster deployment.

POTENTIAL CHALLENGES & RISKS

01 **Connectivity:** Unstable internet in rural/remote areas may affect real-time sync.

02 **Adoption:** Elderly users may take time to adopt new technology.

03 **Data Privacy:** Sensitive health and cognitive data require strict protection.

04 **Accuracy:** AI recommendations depend on quality and consistency of user data.

05 **Resource Constraints:** Limited devices & digital literacy among some caregivers.

STRATEGIES FOR OVERCOMING CHALLENGES

01 Dual-mode sync (Wi-Fi + Offline) and local data storage for uninterrupted usage.

02 Gamified onboarding, voice guidance and caregiver training sessions to boost adoption.

03 End-to-end encryption, role-based access and consent management for data security.

04 Continuous model training, feedback loops and manual review for recommendations.

05 Lightweight app, offline-first design and multi-language support to ensure inclusivity.

KEY ENABLERS

- Secure & Private by Design
- Offline First Architecture
- Caregiver Centric
- Modular & Scalable
- Open Source Technologies
- Affordable & Sustainable

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IMPACT AND BENEFITS

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ELDERLY PERSON



- Personalized cognitive engagement
- Greater independence in daily life
- Familiar, simple and less confusing experience

NER COMMUNITY



- Multilingual and voice-enabled access
- Works in low-connectivity and offline mode
- Promotes inclusive digital elderly care across NER

COGNICARE IMPACT LOOP

A continuous cycle of engagement, adaptation and support for better cognitive well-being.

01 ENGAGE
Engaging games and activities stimulate the mind and collect meaningful signals.

02 ADAPT
AI adapts difficulty and content to each individual in real-time.

03 SUPPORT
Smart reminders, routines and insights provide daily living support.

04 CONNECT
Caregivers stay connected through insights, alerts and progress updates.

05 IMPROVE
Continuous learning improves cognitive abilities and quality of life over time.

CAREGIVERS & FAMILY



- Real-time insights and smart alerts
- Early detection of changes
- Reduced stress and monitoring burden
- Better confidence in care

HEALTHCARE PROFESSIONALS



- Structured longitudinal data for better decisions
- Better visibility between visits
- Early cognitive signals support timely care planning

FROM ONE SESSION TO LONG-TERM COGNITIVE JOURNEY

Game Session
User plays cognitive games/ activities

Performance Signals
System captures behavior, scores and responses

Personal Baseline
AI builds a unique baseline for each individual

AI Detects Change
Detects patterns and flags cognitive changes early

Personalized Support
AI adapts games, reminders and daily support

Caregiver Insight
Caregivers get clear, timely insights and alerts

Long-Term Journey
Continuous improvement in cognition and quality of life

TRANSFORMING LIVES, EVERY DAY

Better Cognitive Engagement
Personalized training helps maintain memory, attention and problem-solving skills.

Independent Living
Smart support and routines help users stay independent for longer.

Empowered Caregivers
Actionable insights and alerts enable caregivers to provide timely support.

Stronger Healthcare Systems
Data-driven insights assist professionals in early intervention and better outcomes.

Inclusive Social Impact
Accessible, affordable and adaptive technology promotes dignity and inclusion for elderly across NER.

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RESEARCH AND REFERENCES

KEY REFERENCES

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- **Alzheimer's Association** – 2024 Alzheimer's Disease Facts and Figures
  https://www.alz.org/alzheimers-dementia/facts-figures
- **NIH** – Research on Cognitive Health & Aging
  https://www.nia.nih.gov/health/cognitive-health
- **Journal of Alzheimer's Disease** – Recent Studies
  https://content.iospress.com/journals/journal-of-alzheimers-disease
- **IEEE Xplore** – IoT & AI in Healthcare
  https://ieeexplore.ieee.org/Xplore/home.jsp
- **PubMed** – Mental Health & Personalized Interventions
  https://pubmed.ncbi.nlm.nih.gov/

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Global health research reports and datasets.
https://www.researchandmarkets.com

Frontiers in Aging Neuroscience
Peer-reviewed articles on neuroscience and dementia.
https://www.frontiersin.org

SciTechDaily
Latest science and technology news on brain health.
https://scitechdaily.com

Nature Aging
High-impact research on aging and cognitive health.
https://www.nature.com/nataging

MDPI Brain Sciences
Open access research on brain and mental health.
https://www.mdpi.com/journal/brainsci

Towards AI
AI advancements and applications in healthcare.
https://towardsai.net

Ministry of Health & Family Welfare
National health policies and programs, India.
https://mohfw.gov.in

ICAR
Research and publications on health & nutrition.
https://icar.org.in

Google Scholar
Scholarly articles across domains and publications.
https://scholar.google.com

OUR APPROACH: We leverage credible research, global reports, and cutting-edge AI/IoT technologies to build an evidence-based, user-centric solution that creates meaningful impact for elderly care.